Rude or Racist?
Distinguishing Hate Speech from Offensive Language Using a Fine-Tuned DistilBERT Model

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Abstract

Language, especially online, is often crude and insulting. Yet, such statements do not always carry hateful intent. To be effective, automated hate speech detection must be able to recognize this intent and differentiate between hate speech and offensive language. As traditional NLP methods struggle with this distinction, this study investigates whether a finetuned DistilBERT model can differentiate hate speech from offensive language better. The model trained detects hate speech (F1 0.74) and non-offensive speech (F1 0.72) decently, but struggles with the identification of offensive language (F1 0.52), resulting in an overall F1 score of 0.67 on the test set. The loss curves suggest that the model underfitted. I find that the finetune approach to hate speech detection is effective, but has limitations concerning resource use that cause detection issues with more implicit forms of derogation.

1 Introduction

Language on social media is often crude and insulting. Yet, such offensive utterances are often emitted without hateful intent (Davidson et al., 2017). The lack of malicious intent shows that this use of language and hate speech should be treated as distinct phenomena, especially as the latter can have consequences linked to real world violence: UN investigators found Facebook to have played a leading role in the genocide of the Rohingya community in Myanmar, by allowing the spread of hate speech on their platform (Mathew et al., 2019). Governments, and some social media sites, are trying to curb the spread of hate speech, but as hateful content spreads faster and farther than non-hateful content, such curbing remains difficult (Mathew et al., 2019).
Effective countermeasures depend on automated hate speech detection, which in turn need to be able to distinguish it from offensive language. Yet much of the existing work conflates the two categories, classifying any offensive content as hate speech and thereby lowering the precision of the resulting models (Davidson et al., 2017). To separate the two categories, Davidson et al. 2017 introduce a definition of hate speech, that deliberately does not include all instances of offensive language, even allowing for certain slurs to be excluded depending on context. The definition, which I will be following for this study, reads:
"[. . .] language that is used to expresses hatred towards a targeted group or is intended to be derogatory, to humiliate, or to insult the members of the group." (Davidson et al., 2017)

Davidson et al. 2017 labeled a social media corpus with such a threefold distinction (hate speech, offensive and neither), and evaluated different classical machine learning classifiers on it. Their best model, a logistic regression, achieved an F1-score of 0.90 overall, but had performance issues within the hate speech class. With only a precision of 0.44 and a recall of 0.61, more than a third of all hate speech goes undetected. Even with a clear distinction between the two categories, their models struggled to tell hate speech from offensive language.
Since the Davidson et al. 2017 study, however, NLP has expanded to new approaches, such as the pretrained language model BERT (Devlin et al., 2019). BERT is a bidirectional transformer model, designed to be easily finetuned for a wide range of tasks (Devlin et al., 2019). In finetuning the pretrained model is equipped with a small classification head and further trained on a labelled dataset for specific tasks like text classification. As the confusion between hate speech and offensive language hinges precisely on contextual and pragmatic cues instead of keywords (Davidson et al., 2017), such pretrained representations are a promising candidate for improving on the results

of Davidson et al. 2017. This raises the research question of the present study: Can a fine-tuned BERT model distinguish hate speech from merely offensive language more accurately than the classical machine learning models?

In this study I find that a finetuned model is able to surpass those classical models in edgecases like the "hate speech" and "neither" category. Yet, without the necessary resources and training data, it fails when it comes to weaker classes, such as the "offensive" category. The main contribution of this study are as follows:

- I conduct a comparative study, investigating how well a finetuned DestilBert model compares to the traditional machine learning appraoch in Davidson et al. 2017.
- An error analysis on the class level.
- A discussion of the models focus on specific keywords over context.

In the following chapter(Chapter 2), I will discuss similar studies, contrasting the newer deep learning approaches with more traditional NLP methods. Continuing to Chapter 3, I will elaborate on the data used in this study and the experimental setup. Following this, Chapter 4 contains the studies results, including a discussion of the models prediction errors. I finish this study in Chapter 5, with the conclusion.

2 Related Work

2.1 Deep learning approaches

Automated hate speech detection has become a prominent task in computational linguistics, with a range of possible classification approaches. In a broader overview, Garrido-Merchan et al. 2023 argue that pretrained models such as BERT should be considered as default candidates for NLP-classification tasks, such as hate speech detection. In their comparisons with traditional models, Garrido-Merchan et al. 2023 report better performance for BERT across all scenarios tested. Within this line of thinking Mozafari et al. 2020 finetune a BERT-model on social media data, for hate speech detection specifically. By introducing a reweighing-based bias alleviation system to the model, they reduce systematic racial bias found in trained classifiers. Bias that is evident in the model's disproportionate assignment of negative labels to posts aligned with African American English (Mozafari et al., 2020). Ousidhoum et al. 2019 extend this task beyond a single language. They introduce a dataset including hate speech in English, French and Arabic and test, among other tasks, multilingual hate speech classification. They find that deep learning approaches outperform traditional machine learning in multilingual classification tasks (Ousidhoum et al., 2019).

2.2 Traditional approaches

Solopova et al. 2023 offer a useful contrast in approach. Within a classification task focused on propaganda, they contrast a multilingual BERT-model with classical machine learning approaches, such as an SVM. They find that the SVM model reached similar results as the BERT-model, making up for its lower F1-score (0.92 to 0.87), with higher explainability (Solopova et al., 2023). While this evidence comes from propaganda detection, it illustrates the need for explainability in classification tasks and its lack in pretrained language models. In line with this need, Malmasi and Zampieri 2017 investigate hate speech detection using an SVM and different n-grams. Their best feature set achieved 78% accuracy, but their error analysis identifies the central difficulty: hate speech and generally offensive language are often confused (Malmasi and Zampieri, 2017). This work builds on the dataset and three-way annotation scheme of Davidson et al. 2017, which made that distinction explicit rather than collapsing all offensive content into a single non-hate category. Davidson et al. 2017 likewise report that the boundary between hate speech and offensive language is challenging, reinforcing the importance of distinctive label definitions and context when interpreting model performance.

While traditional classification task approaches struggle with the distinction between offensive language and hate speech, they are more transparent than their deep learning counterparts. This difference underlines the need for a differentiated discussion, about the loss transparency in favor of higher prediction power.

3 Materials and Methods

3.1 Material

Since the original data of Davidson et al. 2017 has been reported to contain labelling errors (Davidson et al., 2017), I instead opted for the HateXplain corpus from Mathew et al. 2021. It covers the same

annotation scheme (labelling for hate speech, offensive speech and neither) and sources its data from similar social media sites (Mathew et al., 2021). It comprises roughly 20,000 English posts from Twitter (now "X") and Gab, collected in the timespan of 2016 and 2020 (Mathew et al. 2019, Mathew et al. 2021). The corpus is split into training, validation and test set using a 80/10/10 split. Each post was labelled using the crowd working platform Amazon Mechanical Turk. The data is published under the MIT license and is freely available on Huggingface[1]. Mathew et al. 2021 anonymised it by replacing usernames with <user> and excluded posts in other modalities than text. To keep the results comparable with Davidson et al. 2017, I simplified the corpus. Each post received the majority label of its annotators (computed with statistics[2]), and additional columns were dropped. As this study works with hateful language, slurs will be discussed and censored accordingly. Examples for all three categories are:

Hate speech *n∗gger f∗ggot r∗tard b∗aner dishwasher time to see if i get ban*

Offensive speech *he said bro even your texts sound redneck whatever the fuck that means <user>*

Neither *<user> why should you ban immigrants from operating small business every business start somewhere and tomorrow that will give people job*

3.2 Experimental setup

3.2.1 Preprocessing

The training set was then cleaned and preprocessed. Dropped from the data were 23 duplicates, 13 broken entries, detected with ftfy (Speer, 2019), and 263 outliers. These outliers were identified using IQR method on the features TTR and text length. Because both features have heavy-tailed distributions, outliers were identified on the log-transformed values. From the removed outliers, 15 posts were unusually short (two to three words) and 248 very repetitive in nature, resulting in a very high TTR (posts containing the same emoji multiple times). Altogether, 283 entries were dropped. As seen in Figure 1, a comparison of the label distribution before and after the removal showed no meaningful shift.

3.2.2 Implementation

As base model, I used DistilBERT (Sanh et al., 2019): a pretrained general-purpose model that is 40% smaller and 60% faster than BERT while retaining 97% of its language-understanding performance (Sanh et al., 2019). This makes it suitable for studies with limited resources. All code ran on the free tier of Google Colab [3] using a T4 GPU and Python 3.13.15. I utilized the integrated DistilBERT WordPiece tokenizer and padded to a maximum sequence length of 512 tokens. To save resources and reduce the risk of overfitting, most of the model was then frozen. Only the last transformer block and the classification head were trained.

Hyperparameters were optimised with Optuna (Akiba et al., 2019), a Bayesian optimisation framework, and the HuggingFace Trainer (Wolf et al., 2020). The search space was based on the ranges recommended for BERT fine-tuning (Devlin et al., 2019) (learning rate: 2e-5 to 1e-4, number of epochs: 2 to 4, batch size: 16 or 32, weight decay: 1e-4 to 0.1). Because of limited resources, the search was run on a smaller class-stratified subset of 1000 posts. The best trial used the following hyperparameters:

- learning rate of 8.4e-05
- weight decay of 1.5e-3
- a batch size of 16
- 4 epochs

With these hyperparameters, the model was trained on the full training set using the Hugging Face Trainer (Wolf et al., 2020) on PyTorch (Paszke et al., 2019). Optimisation used AdamW (Loshchilov and Hutter, 2017). The complete code is available on GitHub[4].

3.2.3 Evaluation Metrics

The model was evaluated on the validation set, saving the checkpoint with the best F1-score as trained model. The final performance was measured once on the held-out test set, using accuracy

[1] https://huggingface.co/datasets/Hate-speech-CNERG/hatexplain

[2] https://docs.python.org/3/library/statistics.html

[3] https://colab.research.google.com/

[4] https://github.com/zoaltrub/RudeOrRacist_HA_VAMOS

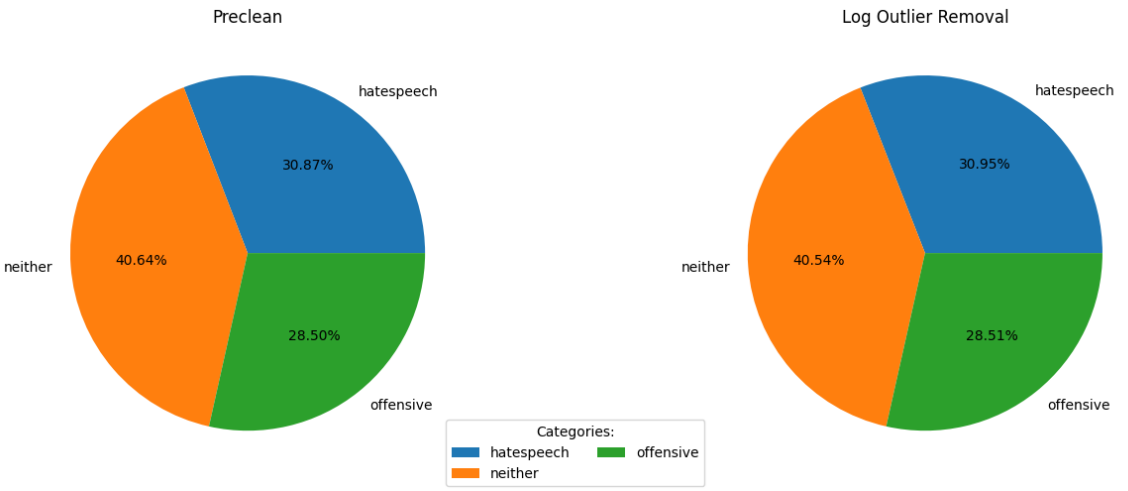


Figure 1: Confusion matrix showing spread of spread of predictions, in comparison to true labels

Metric	Validation Set	Test set
Precision	0.68	0.67
Recall	0.69	0.68
Accuracy	0.69	0.68
F1 score	0.68	0.67

Table 1: Table showing the different metrics of accuracy, precision, recall and F1 on the validation and test set.



Figure 2: Figure showing the loss on the training data throughout the finetuning process.

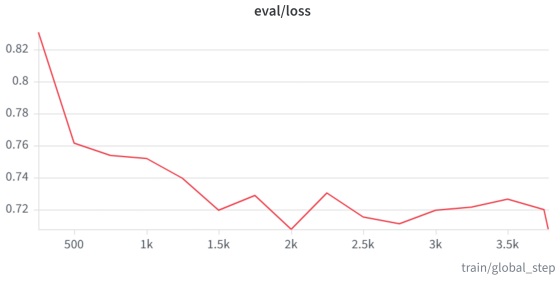


Figure 3: Figure showing the loss on the validation data throughout the finetuning process.

and weighted precision, recall and F1, the latter for comparison with Davidson et al. 2017.

4 Experiment Results

4.1 Performance Analysis

With this configuration the model reached a weighted F1 score of 0.68 on the validation set, with an accuracy of 0.69 and a precision of 0.68. On the held-out test set, the final model achieved an F1 score of 0.67, an accuracy of 0.68 and a precision of 0.67, a difference of about one percentage point (See Table 1).

During finetuning, both training and validation loss behaved similar, decreasing at a comparable pace (Figure 2 and 3). Therefore, the loss curves show no sign of overfitting. The final training loss was 0.60 and the validation loss 0.72, showing that, while the model had learned meaningful patterns, it has underfitted and never properly described the data. This is most likely due to the limited resources of this experiment.

Table 2 shows, that performance differed strongly on a the class level. Hate speech was classified with the highest precision of 0.79, a recall of 0.64 and an F1 score of 0.74. The "neither" class was classified evenly, with a precision, recall and F1 score of 0.73, 0.71 and 0.72. Surprisingly, offensive language was by far the most difficult category, with a precision of 0.49, a recall of 0.57 and an F1 score of 0.52.

4.2 Error Analysis

To understand the issues of the finetuned model, misclassified posts of the set were taken into a qualitative analysis. Posts stemming from the

Metric	Neither	Hate Speech	Offensive Speech
Precision	0.73	0.79	0.49
Recall	0.71	0.64	0.57
F1 score	0.72	0.74	0.52

Table 2: Table showing the different metrics of accuracy, precision, recall and F1 on the class level

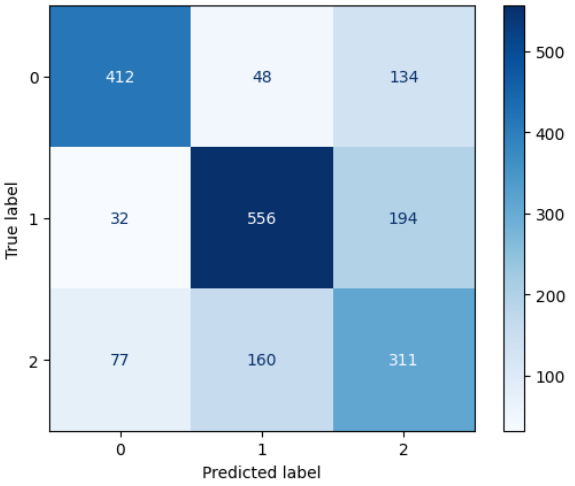


Figure 4: Figure showing the confusion matrix, highlighting spread of prediction results

"neither" category were classified comparatively well, posts falsely assigned to other classes tended to contain more insults and slurs than the correctly classified ones. But this pattern expanded past the "neither" category. Posts from other categories that were falsely assigned "neither" showed less explicit language, while those falsely assigned to "hate speech" contained more. A clear example of this is the case of the n-word. While the more colloquial writing, ending with -a, appeared in all subgroups, the more offensive version (ending -er), was strictly sorted into "hate speech", appearing 23 times in "neither" and 24 times in "offensive" originally. This behaviour suggest that the model strongly relies on semantic content, instead of pragmatic context especially considering examples such as:

Neither, sorted as hate speech: *that video of the deaf dude having sex was killing me yesterday and this n∗gger <user> gon post the video in the chat so i am dying all over again*

Hate speech, sorted as neither: *<user> you are a fake pervert asshole you can be raped dont be stupid and work or study change you life you n*ggas*

Another noticeable trend was the prevalence of sexism in the "offensive" category, that was falsely classified as "hate speech". Many posts that should, following Davidson et al. 2017 definition, be categorized as hate speech, were annotated as "offensive" by the annotators. This trend falls in line with findings from Talat and Hovy 2016, explaining that human coders consider homophobic and racist slurs as hateful, but categorize derogatory speech towards women as merely offensive. One such falsley annotated post reads:

Offensive: *i am a muslim and you whores need to put on a burqa so that someone that is not me accidentally does not stone you to death*

5 Conclusion

In this study I investigated whether a finetuned DistilBERT model can distinguish hate speech from offensive language, better than traditional machine learning models. Hate speech was detected best (F1 = 0.74, precision=0.79 and recall = 0.64), followed by non-offensive language (F1 = 0.72, precision= 0.73 and recall = 0.64), while offensive language remained clearly the most difficult class to classify (F1 = 0.52, precision= 0.49 and recall = 0.57). The model is therefore effective at separating the extreme cases to a certain degree but falls behind in the offensive class, resulting in an overall F1 score of 0.67 on the test set. Compared to Davidson et al. 2017, the results show a mixed picture: hate speech was detected better than in their study (they reached 0.44 for precision, and 0.61 for recall), whereas the overall performance was lower (F1 = 0.90). The error analysis suggests that the transformer takes a different route to a similar problem: both approaches struggle at the boundary between hateful and merely offensive language. In this the errors seem to arise from a strong dependence on lexical and semantic cues, which makes implicit derogation hard to detect and leaves colloquial or ambiguous vocabulary open to misclassification.

Additionally, the training curves and the comparatively high loss indicate that the model underfitted the data. This most likely limited the results. In

the training progress a test model was trained, that points int the same direction: on a small sample, the training loss could be driven close to zero within 50 epochs, so the pipeline was in principle able to fit data. Longer and more intensive training, with less frozen layers and more data, would therefore likely have reduced part of the errors.

Considering other finetuned models, that reached a higher prediction power[5][6][7], I conclude that yes, the finetune approach to hate speech identification is effective, but with limitations. The need for data and resources limits the training, causing problems with more implicit or socially accepted forms of derogation, as seen in the in the "offensive" category. Additionally, due to this lack of resources, I find that this model has too low of a prediction power to make up for the lack of transparency. In the future. more exhaustive studies should discuss this trade-off with more adapted models.

Limitations

While important for a comparison to Davidson et al. 2017, the limitation of the training data, by deleting additional information such as rationals or summarizing the more differentiated labeling system, has hurt the model during the training process. Important to note as well is that the presented model is not as strong as more experienced finetuners would have managed for it to be. This is because, at its heart, this study was an attempt at learning the concept of finetuning. This, in combination with the lack of resources, caused the model to remain weak.

References

Takuya Akiba, Shotaro Sano, Toshihiko Yanase, Takeru Ohta, and Masanori Koyama. 2019. Optuna: A next-generation hyperparameter optimization framework. In *Proceedings of the 25th ACM SIGKDD international conference on knowledge discovery & data mining*, pages 2623–2631.

Thomas Davidson, Dana Warmsley, Michael Macy, and Ingmar Weber. 2017. Automated hate speech detection and the problem of offensive language. In *Proceedings of the international AAAI conference on web and social media*, volume 11, pages 512–515.

Jacob Devlin, Ming-Wei Chang, Kenton Lee, and Kristina Toutanova. 2019. Bert: Pre-training of deep bidirectional transformers for language understanding. In *Proceedings of the 2019 conference of the North American chapter of the association for computational linguistics: human language technologies, volume 1 (long and short papers)*, pages 4171–4186.

Eduardo C Garrido-Merchan, Roberto Gozalo-Brizuela, and Santiago Gonzalez-Carvajal. 2023. Comparing bert against traditional machine learning models in text classification. *Journal of Computational and Cognitive Engineering*, 2(4):352–356.

Ilya Loshchilov and Frank Hutter. 2017. Decoupled weight decay regularization. *arXiv preprint arXiv:1711.05101*.

Shervin Malmasi and Marcos Zampieri. 2017. Detecting hate speech in social media. *arXiv preprint arXiv:1712.06427*.

Binny Mathew, Ritam Dutt, Pawan Goyal, and Animesh Mukherjee. 2019. Spread of hate speech in online social media. In *Proceedings of the 10th ACM conference on web science*, pages 173–182.

Binny Mathew, Punyajoy Saha, Seid Muhie Yimam, Chris Biemann, Pawan Goyal, and Animesh Mukherjee. 2021. Hatexplain: A benchmark dataset for explainable hate speech detection. In *Proceedings of the AAAI conference on artificial intelligence*, volume 35, pages 14867–14875.

Marzieh Mozafari, Reza Farahbakhsh, and Noël Crespi. 2020. Hate speech detection and racial bias mitigation in social media based on bert model. *PloS one*, 15(8):e0237861.

Nedjma Ousidhoum, Zizheng Lin, Hongming Zhang, Yangqiu Song, and Dit-Yan Yeung. 2019. Multilingual and multi-aspect hate speech analysis. In *Proceedings of the 2019 conference on empirical methods in natural language processing and the 9th international joint conference on natural language processing (EMNLP-IJCNLP)*, pages 4675–4684.

Adam Paszke, Sam Gross, Francisco Massa, Adam Lerer, James Bradbury, Gregory Chanan, Trevor Killeen, Zeming Lin, Natalia Gimelshein, Luca Antiga, et al. 2019. Pytorch: An imperative style, high-performance deep learning library. *Advances in neural information processing systems*, 32.

Victor Sanh, Lysandre Debut, Julien Chaumond, and Thomas Wolf. 2019. Distilbert, a distilled version of bert: smaller, faster, cheaper and lighter. *arXiv preprint arXiv:1910.01108*.

Veronika Solopova, Oana-Iuliana Popescu, Christoph Benzmüller, and Tim Landgraf. 2023. Automated multilingual detection of pro-kremlin propaganda in newspapers and telegram posts. *Datenbank-Spektrum*, 23(1):5–14.

Robyn Speer. 2019. ftfy.

[5]https://huggingface.co/TaiwoOgun/minilm-hate-speech-onnx

[6]https://huggingface.co/AuricErgeson/hate-speech-detector

[7]https://huggingface.co/KSvendsen/EA-HS

Zeerak Talat and Dirk Hovy. 2016. Hateful symbols or hateful people? predictive features for hate speech detection on twitter. In *Proceedings of the NAACL student research workshop*, pages 88–93.

Thomas Wolf, Lysandre Debut, Victor Sanh, Julien Chaumond, Clement Delangue, Anthony Moi, Pierric Cistac, Tim Rault, Rémi Louf, Morgan Funtowicz, Joe Davison, Sam Shleifer, Patrick von Platen, Clara Ma, Yacine Jernite, Julien Plu, Canwen Xu, Teven Le Scao, Sylvain Gugger, and 3 others. 2020. Transformers: State-of-the-art natural language processing. In *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing: System Demonstrations*, pages 38–45, Online. Association for Computational Linguistics.

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